

Script-Controlled ACL – Restrict Record Access Based on Field Value

5. Project Development Phase

5.1 Database Development

Create User and Record tables. Add the fields required for the access decision, such as Department, Owner_ID, Branch, Status, or Confidentiality_Level.

5.2 User Authentication

Implement login and session management. After successful login, store the authenticated user's identifier and relevant access attributes.

5.3 Record Access Controller

Before a record is returned or changed, the controller receives the user and record information and calls the ACL script.

5.4 Sample Pseudocode

```
FUNCTION checkAccess(user, record, action): IF user.role = 'ADMIN' THEN RETURN ALLOW; IF user.department = record.department THEN RETURN ALLOW; RETURN DENY; END FUNCTION.
```

5.5 CRUD Protection

- Read: return only records that pass the ACL check.
- Create: validate whether the user can create records for the selected field value.
- Update: check access before changing an existing record.
- Delete: check access before deleting a record.

5.6 Error Handling

When access is denied, the application should show a simple authorization message and must not expose restricted record information.

5.7 Development Result

The completed module provides field-value based record protection through programmable ACL rules.